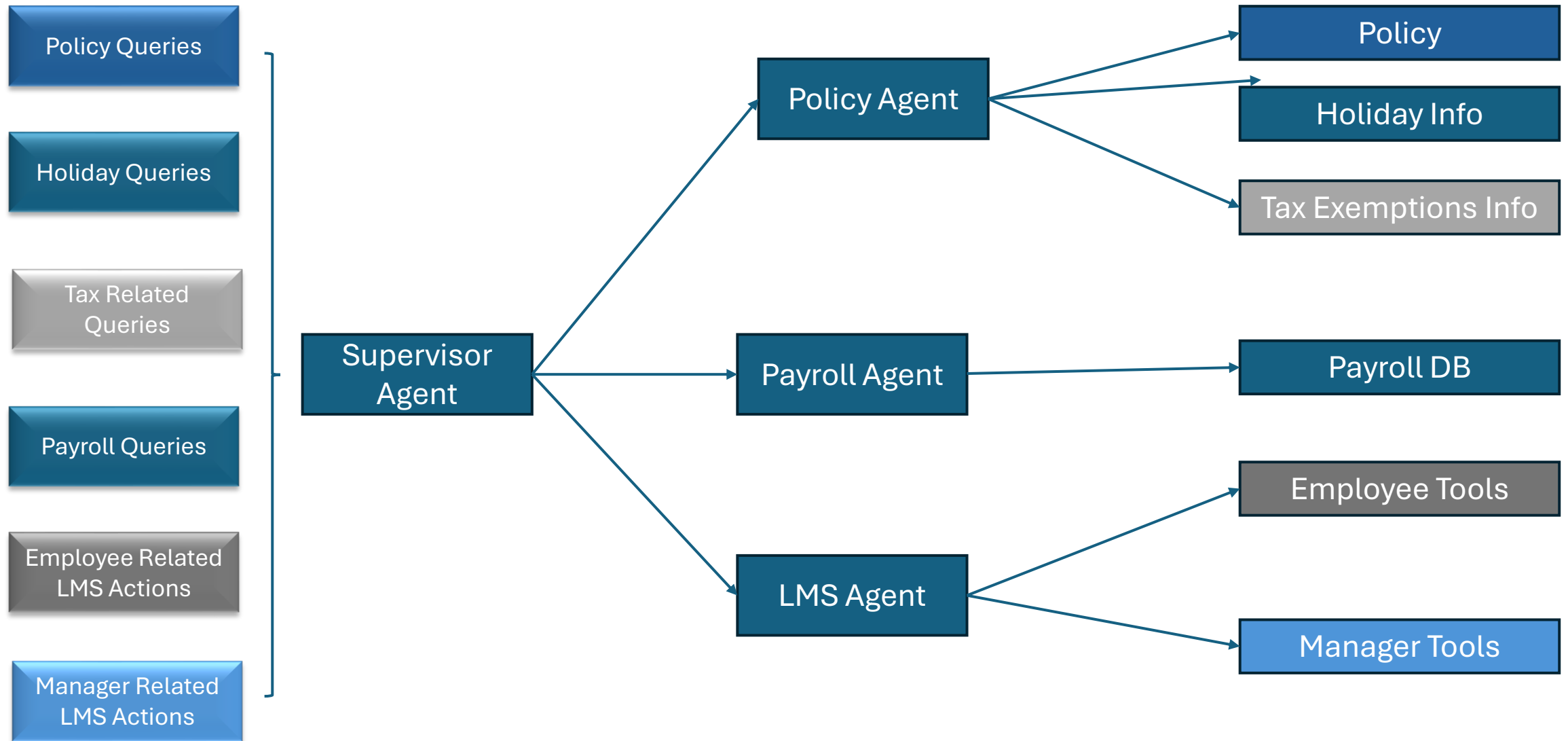




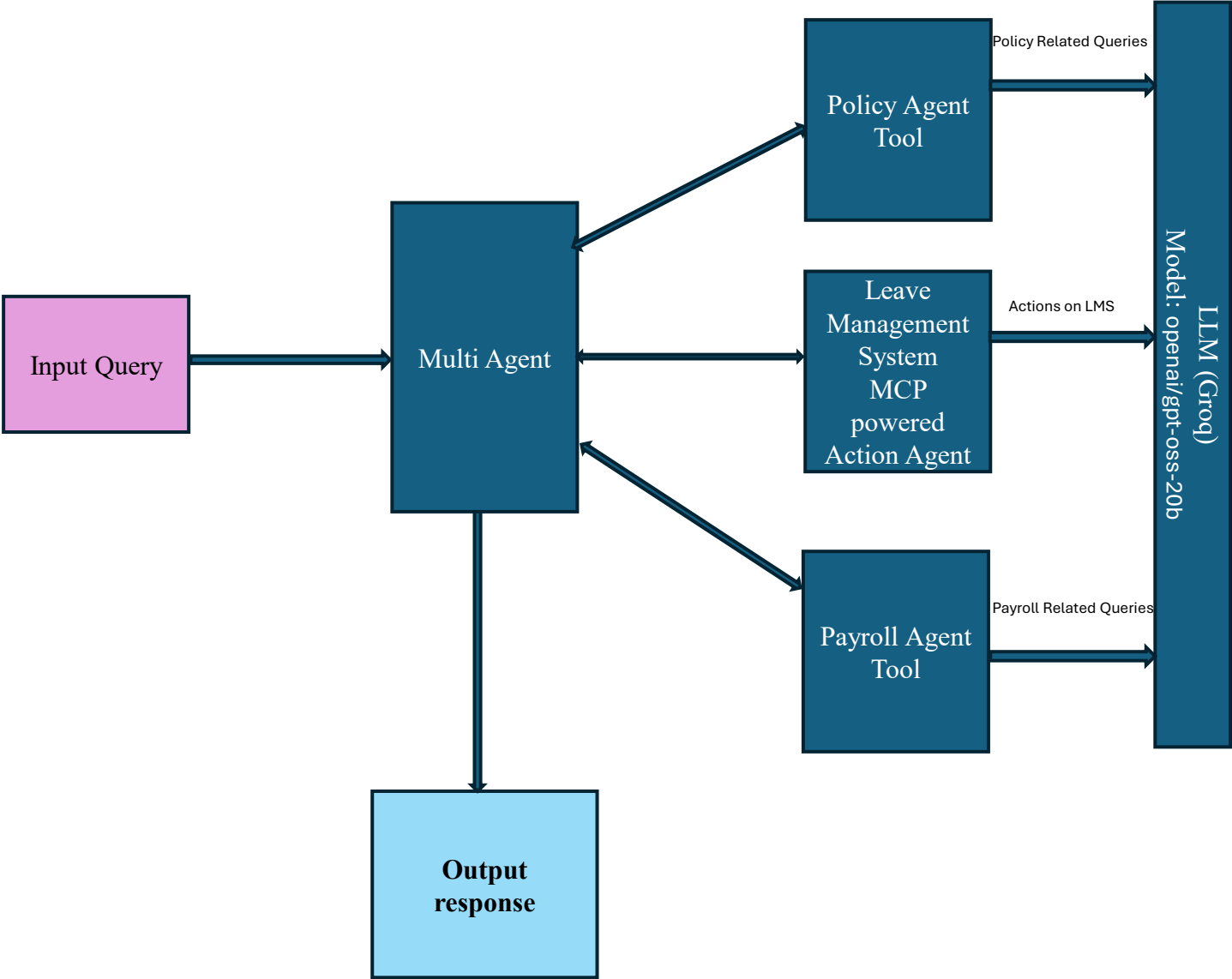
Introduction & Problem Statement:

- HR teams spend significant time on repetitive queries:
 - HR policies related questions
 - Reimbursement policies
 - Resolving Payroll related queries
 - Applying Leave and Checking the balanced leaves
- Employees face delays in getting information.
- Need for AI agent that automates support, reduces workload, and improves employee experience.
- Manual HR support → slow, resource-intensive.
- **Goal:** Build an agentic AI system that can understand queries & Retrieve relevant information from employee data.

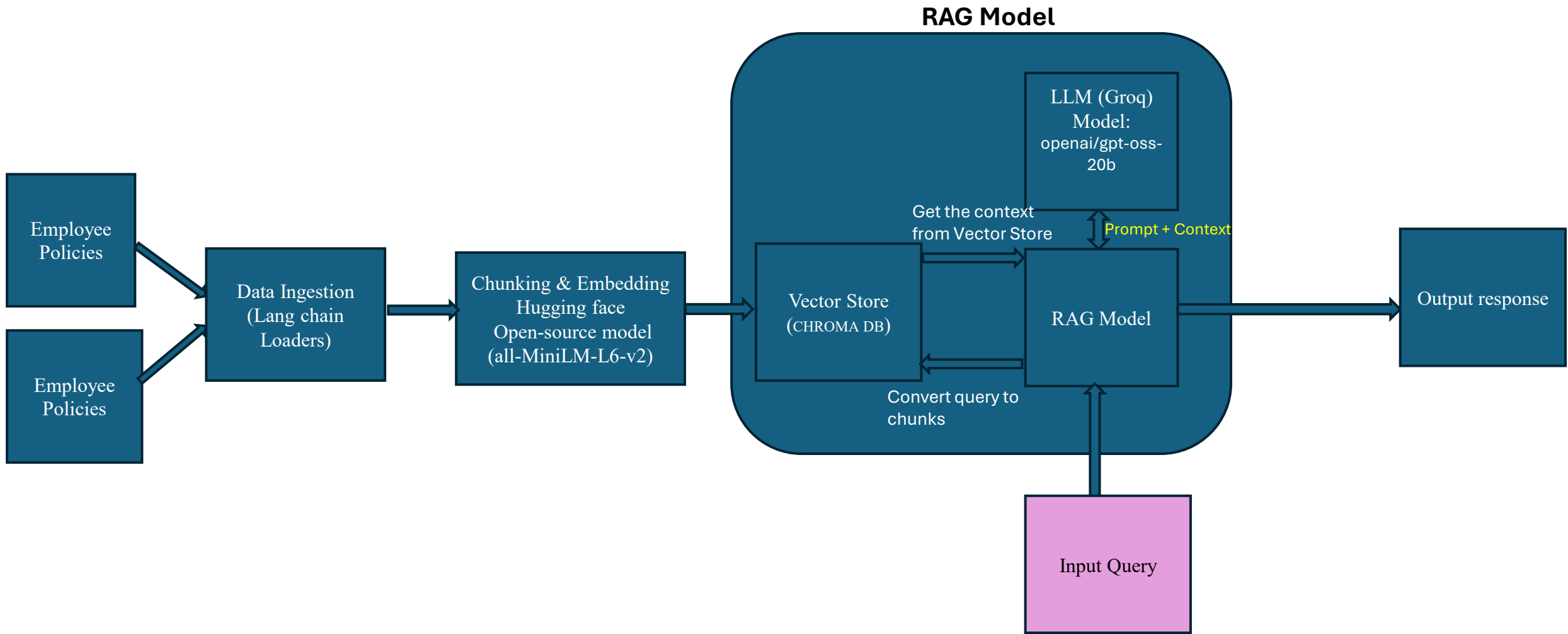
Flow of the Agent:



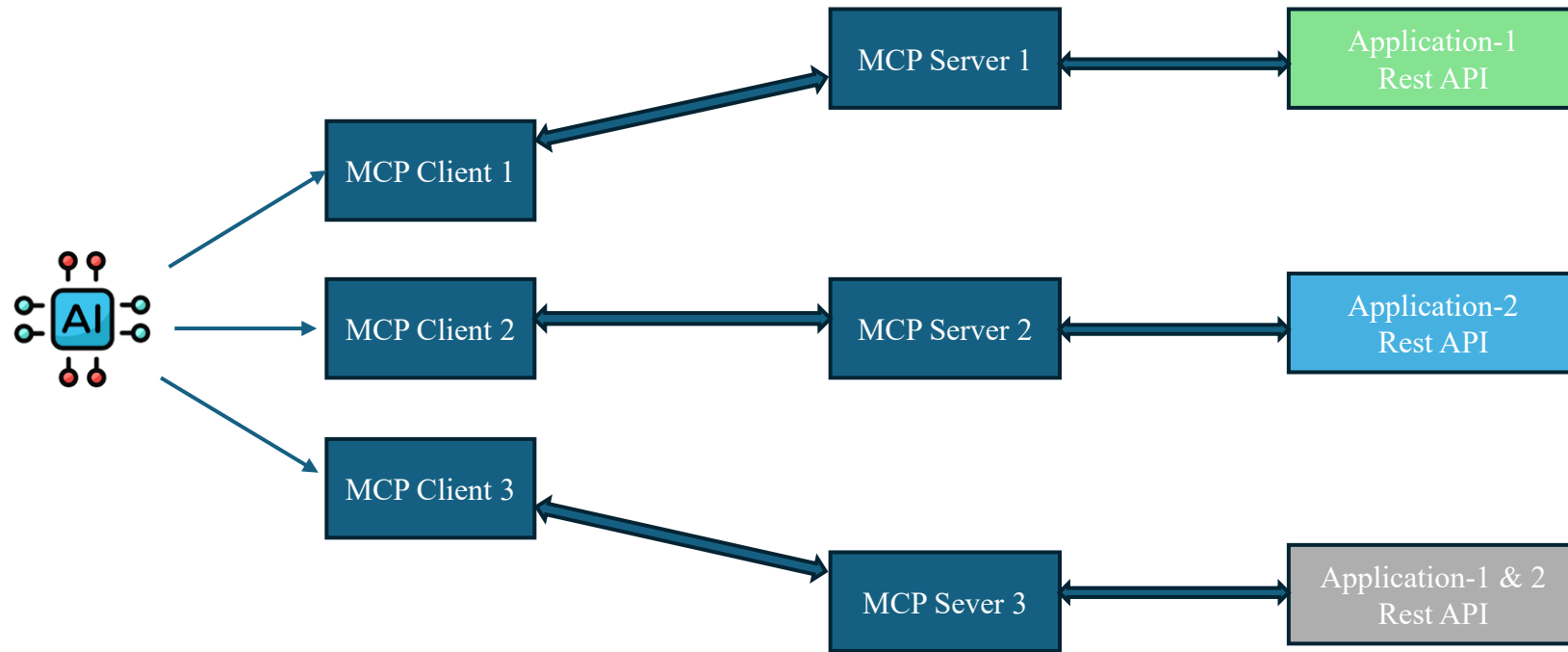
Multi Agent Architecture: (Phase-2)



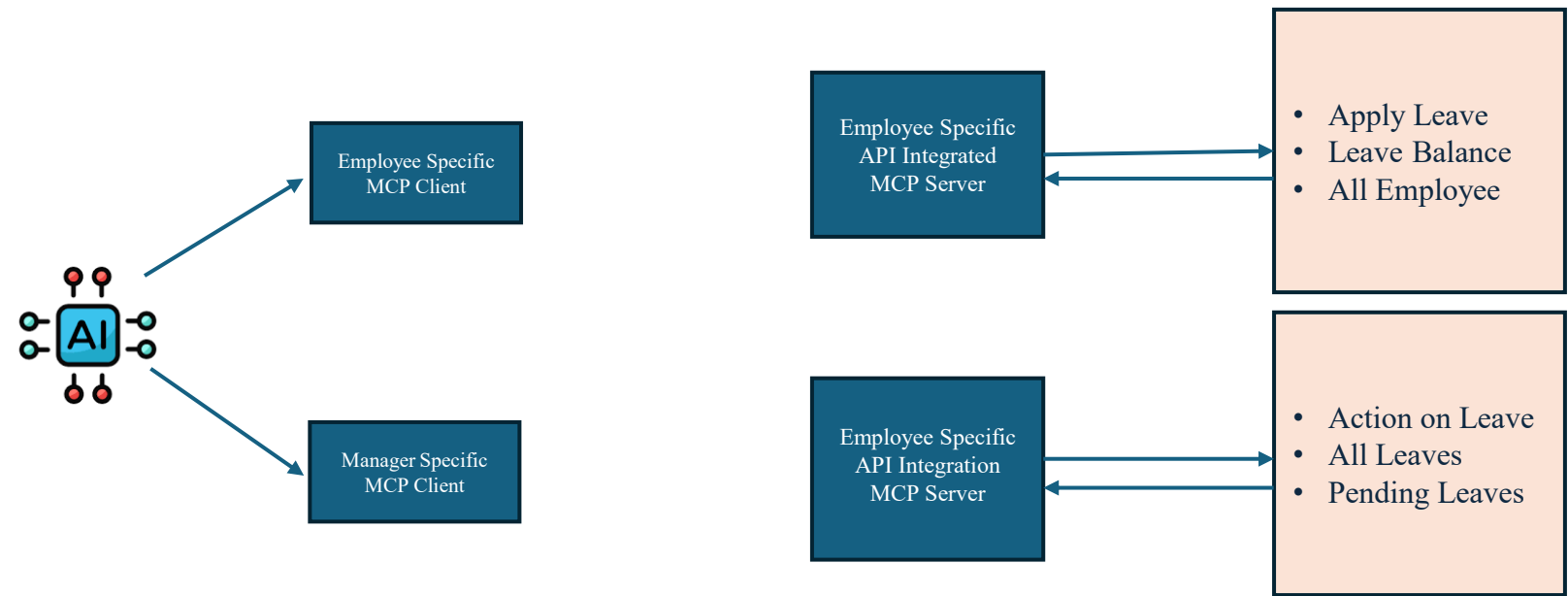
RAG Model:



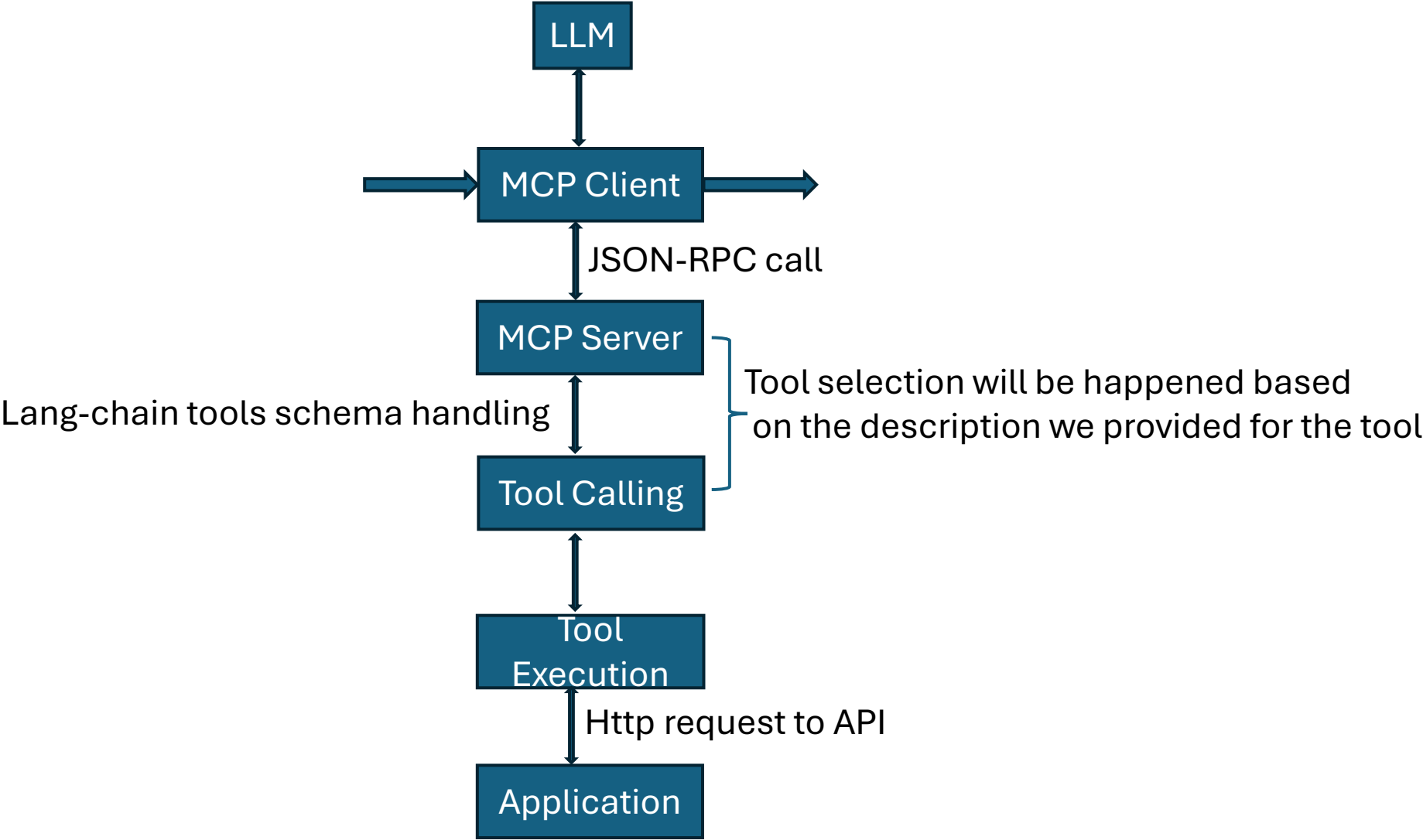
MCP (Modal Context Protocol) Client-Server Architecture:



HR Agent using MCP Modal:



MCP Agent Flow:





Data Requirement :

- HR Documents: Leave policy, reimbursement rules, holiday list.
- Holiday Calendar: JSON/CSV format with official holidays.
- Employee Data (anonymized for testing):
- Leave balance records
- Reimbursement requests

Tools using currently :

- NLP / LLMs: Grok-openai/gpt-oss-20b & GPT-4o(open-source).
- Agent Framework: LangChain
- Context Search: Sentence-transformers/all-MiniLM-l6-v2.
- Database: MySQL
- Frontend: Streamlit

Agent Evaluation:

- Agent Evaluation is done based on the pre-defined questions and scoring for faithful answers
- The pre-defined questions and answers are loaded from the JSON file and comparing the results of questions with actual answers. Based on the matching criteria taking

```
Score:5
Running policy Agent to verify question What will occur if a leave request remains unapproved 60 days after application?
--- Triggered Policy Agent ---
Score:5
Running policy Agent to verify question What is the protocol for applying for Loss of Pay (LOP) leave?
--- Triggered Policy Agent ---
Score:5
Running policy Agent to verify question What might happen if a Line Manager fails to approve exceptional leave requests through the proper channels?
--- Triggered Policy Agent ---
Score:3
Running policy Agent to verify question Are contract employees eligible for Privilege Leave as mentioned in the policy?
--- Triggered Policy Agent ---
Score:5
Agent Evaluation Complete! Success Rate: 75.0%
```

Agent Responses:

Supporting Information

System Status: Online Model: HR-Specialist-v1

Notes:

- This agent has access to employee Payroll, LMS and Policies data.
- You can apply leave from the agent by asking about leave balance or applying for leave.
- Response times may vary based on query complexity.

Clear Chat History

Deploy

HR AI Assistant for Hospital Staff

Assisting hospital staff with HR and Policy inquiries.

what is the Salary of 'Neha Joshi'

Can you apply leave for her for 2 days starting from Mar19th,2026

Casual leave and Reason is for family time.

The net salary for Neha Joshi is \$3,865.00.

Sure! I just need a couple of details to process Neha's leave:

1. What type of leave would you like to apply for? (e.g., sick, casual, annual)

2. Could you share a brief reason for the leave?

Once I have that, I'll submit the request for **Neha Joshi** (Employee ID 4) from 2026-03-19 to 2026-03-20.

Ask me anything about hospital policy...

16-05-2026

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Agent Responses:

<<

Supporting Information

System Status: Online Model: HR-Specialist-v1

Notes:

- This agent has access to employee Payroll, LMS and Policies data.
- You can apply leave from the agent by asking about leave balance or applying for leave.
- Response times may vary based on query complexity.

Clear Chat History

Once I have that, I'll submit the request for Neha Joshi (Employee ID 4) from 2026-03-19 to 2026-03-20.

✔ Your leave request has been submitted successfully!

- Employee ID: 4
- Leave Type: Casual
- Dates: 2026-03-19 to 2026-03-20
- Reason: family time

The request is currently **PENDING** approval. I'll let you know once it's reviewed. If you need anything else, just let me know!

Leave applications for Neha Joshi (Employee ID 4)

Leave ID	Type	Start Date	End Date	Reason	Status
3	Casual	2026-03-16	2026-03-17	Family time	PENDING
4	Casual	2026-03-19	2026-03-20	family time	PENDING

Let me know if you'd like more details or need to take any action.

Ask me anything about hospital policy...

>

Deploy

⋮

Casual leave and Reason is for family time.

what ar the applied for Neha Joshi

16-05-2026

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Expected Outcomes:

- Scalable to other domains (Finance, IT Helpdesk).
- Employees can query HR assistant anytime.
- Automated responses reduce HR workload by 30–40%.
- Personalized responses based on memory of past queries.

References:

- HR-MultiWOZ: A Task Oriented Dialogue (TOD) Dataset for HR LLM Agent (<https://arxiv.org/html/2410.11239v1>)
- Model Context Protocol (MCP) Tool Descriptions Are Smelly! Towards Improving AI Agent Efficiency with Augmented MCP Tool Descriptions (<https://arxiv.org/html/2602.14878v1>)
- Integrating AI and Large Language Models for Automated Data Quality Enhancement in Data Integration Systems (<https://discovery.researcher.life/article/integrating-ai-and-large-language-models-for-automated-data-quality-enhancement-in-data-integration-systems/bf2746490e803ed3a6747d21c6b21f6c>)



*Thank
you!*